Synapsers

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Problem Statement

Decades of valuable study protocols exist, but most of them are trapped in static documents, delaying life-saving treatments. The root cause is that study protocols are non-standardized and unstructured documents, preventing automated analysis.

Key Challenges:

- **Locked Knowledge:** Valuable study protocols are locked in static documents, making them impossible to search, compare, or reuse.
- **Slow & Costly:** Data must be manually extracted, which is inefficient and drives up costs.
- **High Error Rate:** This reliance on manual processes increases the risk of costly errors.

Our Vision: We're transforming every protocol into a living, machine-actionable asset that creates the foundation for a fully automated clinical trial ecosystem.

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Why This Problem?

Market Relevance

Massive Market Inefficiency: Targets a multi-billion dollar clinical research market.

Direct User Pain Point: Solves the daily frustration of clinical teams with slow, manual processes.

High Growth Potential: The industry is actively seeking AI automation for this critical process.

User Impact

Affects Professionals: This frustrates skilled professionals like Clinical Operations Managers, Medical Writers.

Hinders Their Mission: They are trapped by manual processes, preventing them from focusing on advancing science.

Impact on Patients: This ultimately slows the delivery of new treatments to patients.

Business Opportunity

Cost Reduction: Automates manual work to reduce operational spending.

Accelerated Revenue: Faster trials lead to a quicker time-to-market.

New Market Creation: Our platform creates a new ecosystem of automated services, placing us at the center of the digital trial revolution.

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How It Would Help?

Quantifiable Benefits

Up to 90% Faster: Reduce protocol processing from weeks to just a few hours.

Over 85% Labor Cost Reduction: Free up skilled experts from low-value data entry.

10x Downstream Productivity: Structured data accelerates every dependent activity.

>95% Error Elimination: Prevent costly rework with AI.

Scalability

Create a Digital Asset: We create a standardized, machine-readable protocol as the single truth source.

Unlock AI Potential: The standardized output is designed to enable other AI models to automate subsequent tasks.

Build an Ecosystem: We are creating a foundation for AI products that will automate the entire clinical lifecycle.

Strategic Advantage

Solve a Unique Problem: We focus on the critical and complex conversion of protocols into a machine-readable asset.

Empower Users: Our solution empowers medical writers to standardize protocols without expensive programmers.

Gateway to Future AI: We provide the essential bridge that clients need to adopt future AI solutions.

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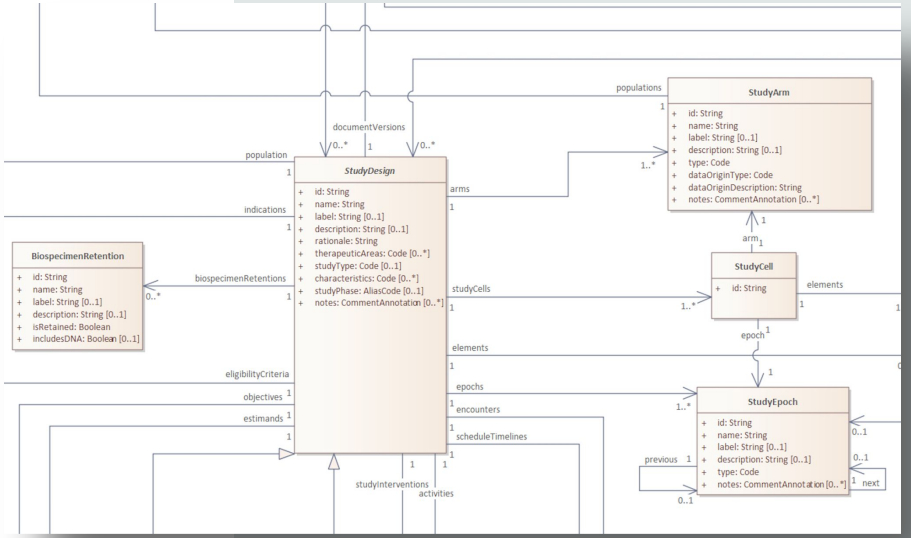
Our approach and how we solved the problem

We are building our solution on the Unified Study Definition Model (USDM), a Industry Standard from CDISC organisation.

USDM:

- The logical data model for study definitions.
- It provides the structured classes and attributes needed to represent study data consistently.
- It contains 85 classes each representing the content from study protocol document.
- It serves as the semantic backbone for a machine-actionable protocol.

USDM



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Our approach and how we solved the problem

- **Ingest & Analyze** - We start by taking any protocol document, in pdf format. Our system first analyzes its structure and parses the raw text.
- **AI-Powered Extraction & Validation** - Our AI engine reads the document and intelligently extracts all critical information required for downstream processes.
    - Human Checkpoint: The user is then prompted to review and approve the extracted content, ensuring complete accuracy from the start.
- **Map to the Industry Standard** - Once validated, the AI maps the data to the CDISC USDM (Unified Study Definition Model)—a globally recognized, vendor-neutral standard for defining clinical studies.
- **Generate the Final Digital Protocol** - Finally, the system connects all the standardized components into a single, unified digital protocol—a portable, reliable, and AI-ready asset.

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The technology stack & architecture

Services:

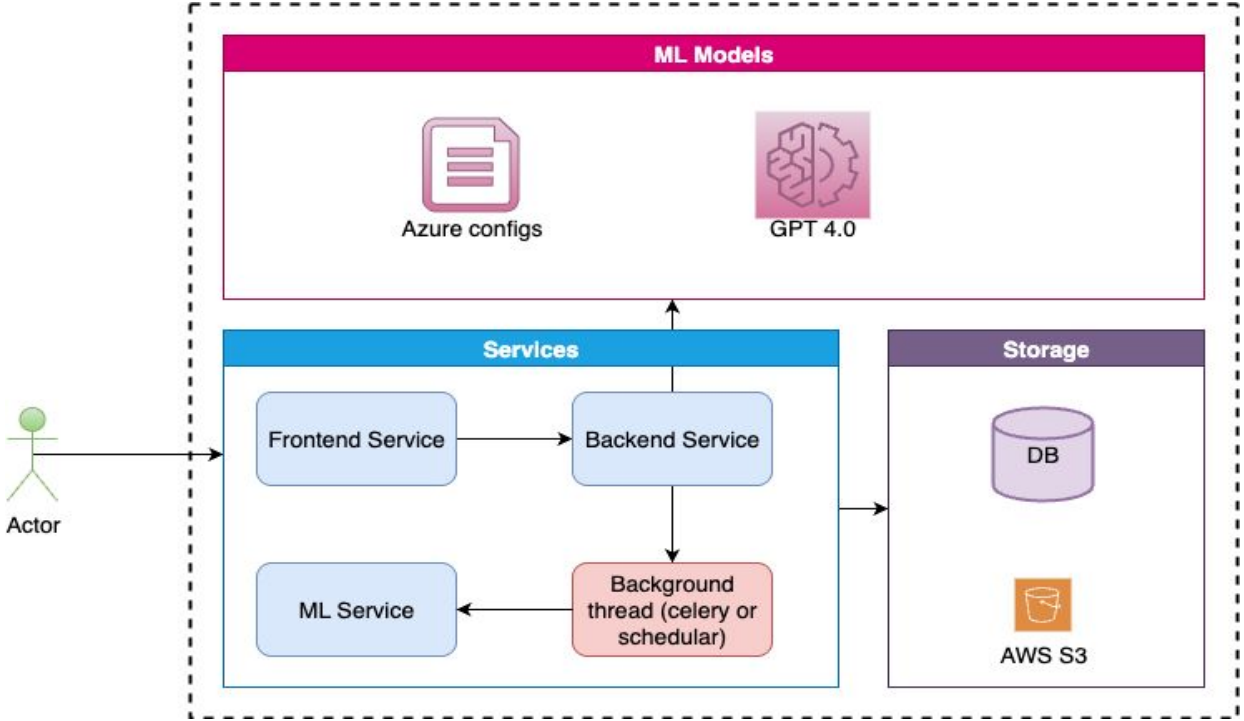
- Python FastAPI
- React Js

Storages:

- Postgres
- S3 bucket

Models:

- GPT 4.0



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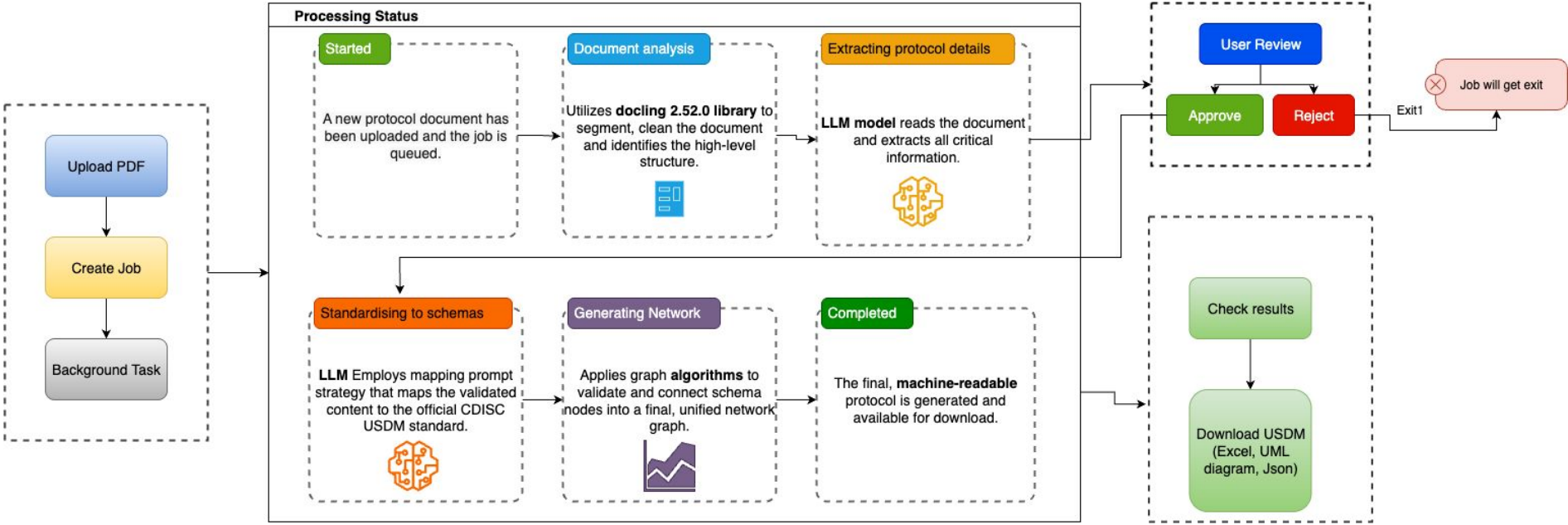
The overall process flow



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Unlocking Downstream Tasks

The digital protocol is the key to unlocking true end-to-end automation. Here's how:

Study design

Automate Study Deployment and Data Setup:

- Protocol Authoring
- Feasibility and Cost

Study start-up

Streamline Changes and Support Subjects:

- Deployment
- Data Collection

Study execution

Automate Statistical Analysis and Data Reporting:

- Amendments
- Subject Information

Analysis Reporting

Simplify Submission Preparation and Review:

- Analysis Automation
- Statistical Analysis Plan (SAP)

Regulatory submission

Support Protocol Authoring to Stakeholder Views:

- Regulatory Review
- Safety Review

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Thank You

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